



Protokoll 2

$$R_{ges} = R_1 + R_{2-6}$$

$$1) R_{3,5} = R_3 + R_5 = (4,7k + 1k)\Omega = 5,7k\Omega$$

$$2) R_{2,3,5} = \frac{R_2 \cdot R_{3,5}}{R_2 + R_{3,5}} = \frac{470\Omega \cdot 5700\Omega}{470\Omega + 5700\Omega} = 434,20\Omega$$

$$3) R_{2,3,5,6} = R_{2,3,5} + R_6 = 434,20\Omega + 220\Omega = 654,2\Omega$$

$$4) R_{2-6} = \frac{R_{2,3,5,6} \cdot R_4}{R_{2,3,5,6} + R_4} = \frac{654,2\Omega \cdot 10000\Omega}{654,2\Omega + 10000\Omega} = 614,03\Omega$$

$$5) R_{ges} = R_1 + R_{2-6} = 330\Omega + 614,03\Omega = 944,03\Omega$$

$$6) I_{ges} = \frac{U_{ges}}{R_{ges}} = \frac{10V}{944,03\Omega} = 10,6\text{ mA}$$

$$7) I_1 = I_{ges} \Rightarrow I_1 = 10,6\text{ mA}$$

$$U_1 = I_1 \cdot R_1 \Rightarrow U_1 = 10,6\text{ mA} \cdot 330\Omega = 3,5V$$

$$8) U_4 = U_B - U_1 = 10V - 3,5V = 6,5V \text{ (Maschensatz)}$$

$$9) I_4 = \frac{U_4}{R_4} = \frac{6,5V}{10k\Omega} = 0,65\text{ mA}$$

$$10) I_{ges} = I_4 + I_6 \Rightarrow I_6 = I_{ges} - I_4 = 10,6\text{ mA} - 0,65\text{ mA}$$
$$I_6 = 9,95\text{ mA}$$

$$11) U_6 = I_6 \cdot R_6 = 9,95\text{ mA} \cdot 220\Omega = 2,2V$$

$$12) U_2 + U_6 = 6,5V \rightarrow U_2 = 6,5V - U_6 = (6,5 - 2,2)V$$
$$U_2 = 4,3V$$

$$13) I_2 = \frac{U_2}{R_2} \rightarrow I_2 = \frac{4,3V}{470\Omega} = 9,2mA$$

$$14) I_6 = I_2 + I_3 \Rightarrow I_3 = I_6 - I_2$$

$$I_3 = 9,95mA - 9,2mA = 0,75mA$$

$$I_5 = I_3 = 0,75mA$$

$$15) U_3 = I_3 \cdot R_3 = 0,75mA \cdot 4,7k\Omega = 3,53V$$

$$U_5 = I_5 \cdot R_5 = 0,75mA \cdot 1k\Omega = 0,75V$$